ECE 370/590 Laboratory Exercise 1: The Dipole Antenna

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[date]

1 Analytical Design (Exercise 1.1)

At $f = 1\,\text{GHz}$,

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8\,\text{m/s}}{1 \times 10^9\,\text{Hz}} = 299.8\,\text{mm}, \qquad \frac{\lambda}{2} = 149.9\,\text{mm}. \tag{1}$$

Table 1: Initial design parameters.

Parameter	Symbol	Value
Total length (tip to tip)	L	150 mm
Conductor	—	Copper tape on foam board
Strip width	w	25.4 mm
Tape thickness	t	0.05 mm
Feed gap	g	12.7 mm

Question 1. What initial dimensions would you select for a dipole resonant near 1 GHz and why?

Answer. Given that the wavelength of a 1 GHz signal is 299.8 mm, a half-wave dipole would ideally have a length of 149.9 mm. The CST simulation we used had a feed gap of 0.5 in and assumed a strip width of 1 in, so we built the antenna to be as representative of the simulated model as possible.

Question 2. What input resistance do you expect at resonance, and how does this compare with the 50 Ω impedance of the VNA?

Answer. At resonance, we expect the input impedance of a half-wave dipole to be about 73Ω. Since the VNA has an impedance of 50 Ω, we expect a slight mismatch at the interface.

Question 3. How should the antenna impedance change as the length is increased or decreased?

Answer. As we make the antenna longer, we expect the resonant frequency to decrease, meaning that we expect an impedance of 73 Ω at a lower frequency. Therefore, the input impedance at 1 GHz should increase as we increase the length of the antenna.

Question 4. How might the conductor diameter or width affect resonance and bandwidth?

Answer. *[Answer here.]*

2 CST Simulation (Exercise 1.2)

2.1 Model setup

We modeled the copper tape dipole *[Solver (time domain), frequency range 0.7 GHz to 1.3 GHz, open (add space) boundaries, discrete 50 Ω port across the feed gap, conductor material (copper vs. PEC), foam board included or not, mesh cell count and convergence check. State the simulated length L.]*

2.2 Reflection coefficient and input impedance

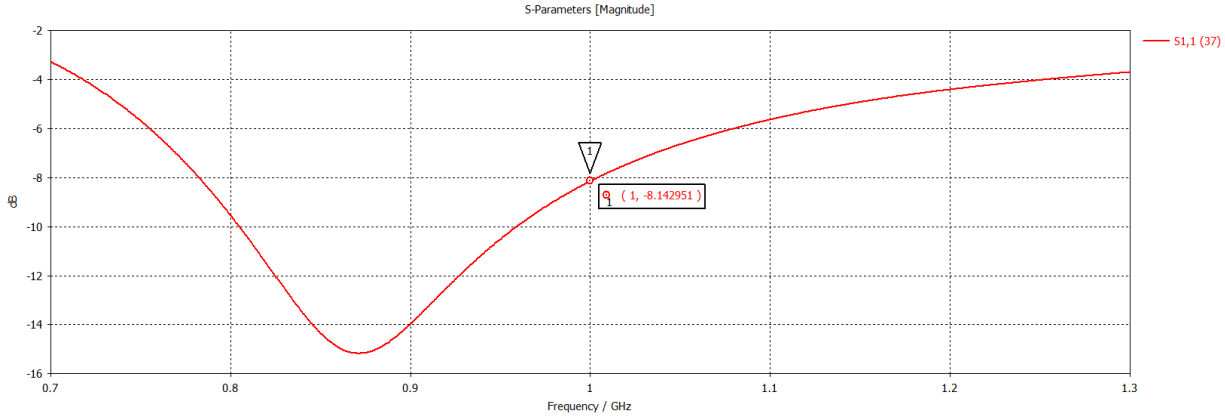


Figure 1: Simulated $|S_{11}|$ (dB) re 50 Ω, 1 in-wide tape dipole. Minimum −15.1 dB near 0.873 GHz; −8.14 dB at 1 GHz.

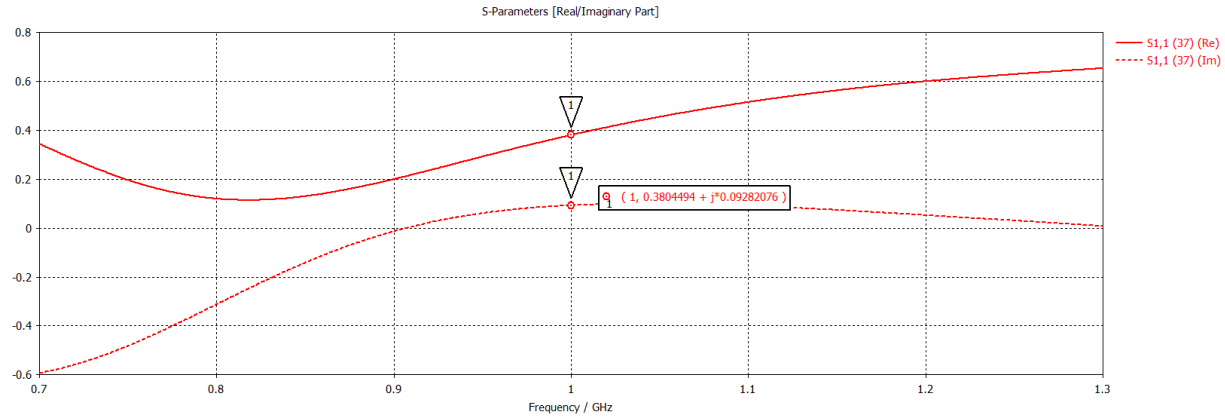


Figure 2: Simulated S_{11}, real and imaginary parts. At 1 GHz: $S_{11} = 0.380 + j0.093$.

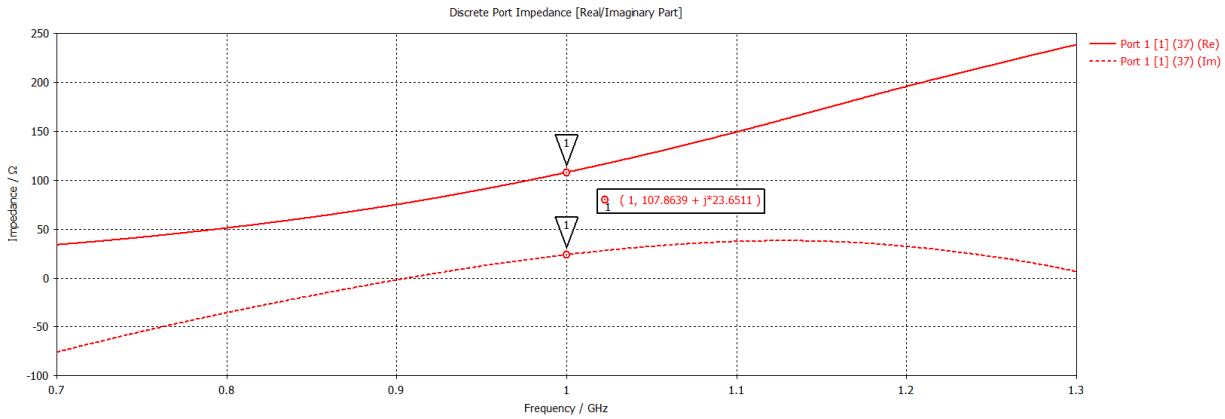


Figure 3: Simulated input impedance Z_{11}. At 1 GHz: $Z_{11} = 107.9 + j23.7\ \Omega$. Reactance crosses zero near *[≈0.90 GHz; read exact value from a marker]*.

Table 2: Simulated resonance summary (1 in tape, L = *[...]*).

Quantity	Value
Zero-reactance frequency $f_{X=0}$	*[≈0.90 GHz]*
R_{in} at $f_{X=0}$	*[≈75 Ω]*
Frequency of min $\lvert S_{11}\rvert$	*[0.873 GHz]*
Min $\lvert S_{11}\rvert$	−15.1 dB
−10 dB band	*[≈0.82 GHz to 0.93 GHz]*
Fractional bandwidth	*[...]*
Z_{11} at 1 GHz	$107.9 + j23.7\ \Omega$
Radiation efficiency at 1 GHz	0.9998
Directivity at 1 GHz	2.19 dBi

2.3 Surface current and far field

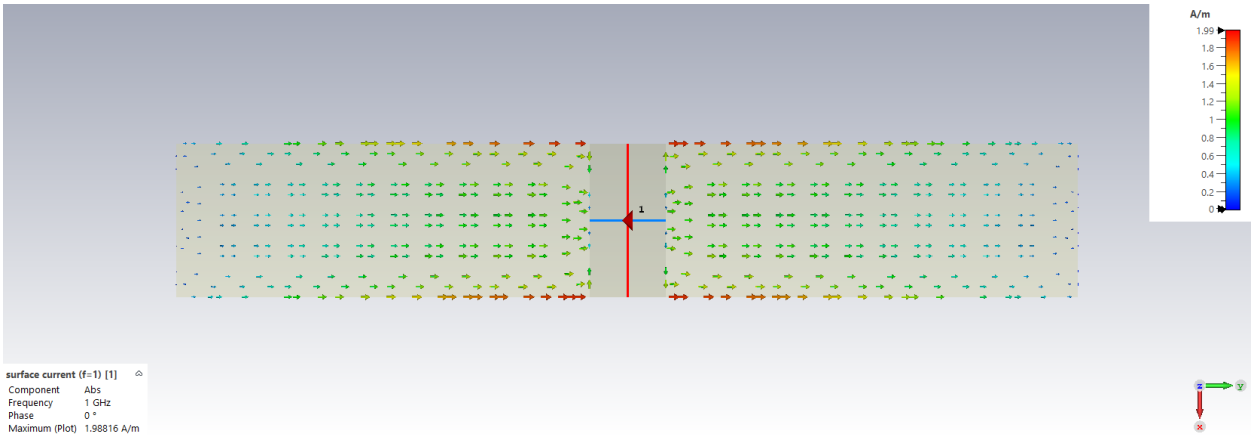
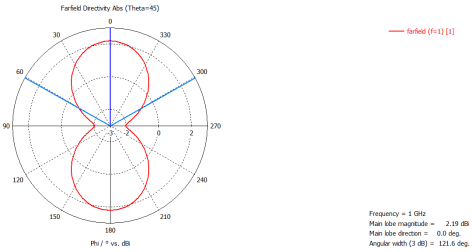
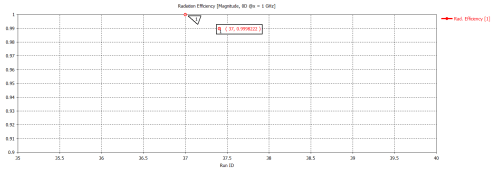


Figure 4: Simulated surface current magnitude at 1 GHz (max 1.99 A/m).



(a) Directivity cut at 1 GHz (main lobe 2.19 dBi, 3 dB width 121.6°).



(b) Radiation efficiency at 1 GHz: 0.9998.

Figure 5: Simulated far-field and efficiency.

2.4 Parameter sweep

[Length sweep for the 1 in model. Only the 0.5 in model currently has a sweep (`Plots/0.5 Inch/`, files `S1,1_130` to `S1,1_160`); either run the same sweep at 1 in or include the 0.5 in sweep and say so.]

> **Question 5.** How closely does the simulated resonant length agree with your analytical estimate?

Answer. *[Answer here.]*

> **Question 6.** What happens to the input impedance as the dipole length changes?

Answer. *[Answer here.]*

> **Question 7.** How does the predicted bandwidth depend on conductor thickness or width?

Answer. *[Answer here.]*

> **Question 8.** What current distribution is observed along the antenna near resonance?

Answer. *[Answer here.]*

Selected fabrication geometry: *[...]*

3 Fabrication (Exercise 1.3)

[Copper tape (1 in wide) on foam board; feed gap; SMA-to-screw-terminal adapter attachment; strain relief; how lengths were changed (trimmed one specimen from 18.5 cm downward, or separate specimens).]

Table 3: Design vs. as-built dimensions.

Dimension	Design	As built
Total length L	*[...]*	18.5 cm, 18 cm, 17 cm, 16 cm and 15 cm
Strip width w	25.4 mm	*[...]*
Feed gap g	*[...]*	*[...]*
Lead length	*[...]*	*[...]*

4 Calibration and Measurement (Exercise 1.4)

4.1 Calibration and reference plane

[LibreVNA SOL calibration over 0.5 GHz to 1.5 GHz (2 MHz steps, 501 points); reference plane at the end of the SMA test cable; adapter and leads not de-embedded (justify). Cable-movement diagnostic result.]

4.2 Measured S_{11}

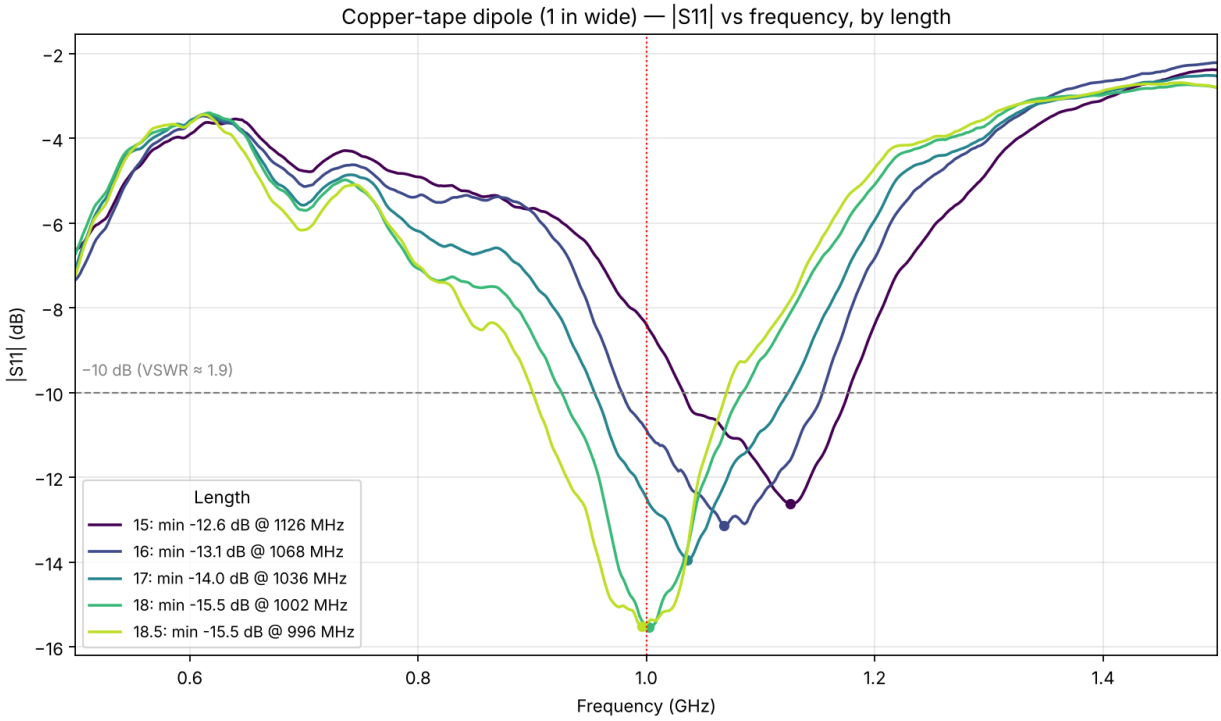
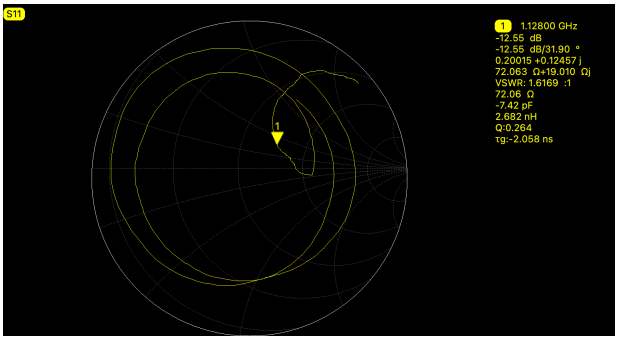
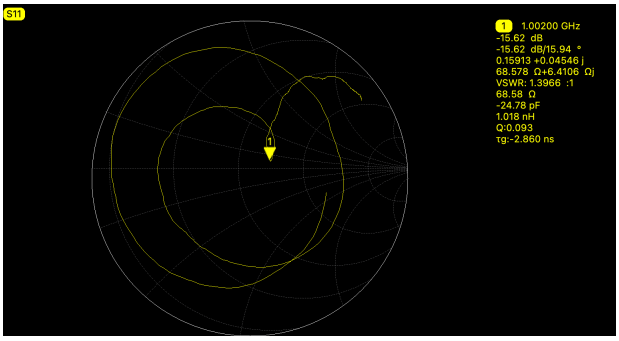


Figure 6: Measured $|S_{11}|$ of the 1 in copper-tape dipole for each length.



(a) Smith chart, $L = 15\,\text{cm}$.



(b) Smith chart, $L = 18\,\text{cm}$.

Figure 7: LibreVNA Smith charts at the shortest and near-resonant lengths. Marker at min $|S_{11}|$.

The input impedance at each marker follows from

$$Z_{\text{in}} = 50\,\frac{1 + S_{11}}{1 - S_{11}}\ \Omega. \tag{2}$$

Table 4: Measured results by length. Minimum $|S_{11}|$ and $-10\,\text{dB}$ band from the CSV exports; Z_{in} from the LibreVNA marker at (or near) the minimum.

L (cm)	$f_{\min\,\lvert S_{11}\rvert}$ (MHz)	min $\lvert S_{11}\rvert$ (dB)	$-10\,\text{dB}$ band (MHz)	BW (%)	Z_{in} at marker (Ω)
15	1126	−12.62	1034–1176	12.6	$72.1 + j19.0$ @ 1128
16	1068	−13.14	980–1154	16.3	$70.2 + j17.7$ @ 1088
17	1036	−13.95	956–1124	16.2	$72.0 + j12.2$ @ 1038
18	1002	−15.53	926–1082	15.6	$68.4 + j6.4$ @ 1002
18.5	996	−15.52	902–1070	16.9	$65.5 + j12.0$ @ 1000

[Zero-reactance frequency for each length. The CSVs hold magnitude only, so $X = 0$ cannot be found from them; re-export complex S_{11} (Touchstone `.s1p`) or read it from the Smith chart.]

5 Geometry Study (Exercise 1.5)

[Resonance shift vs. length, measured vs. CST. Compare slope $\Delta f/\Delta L$. Optional: quarter-wave (1 in vs. 0.5 in) result below.]

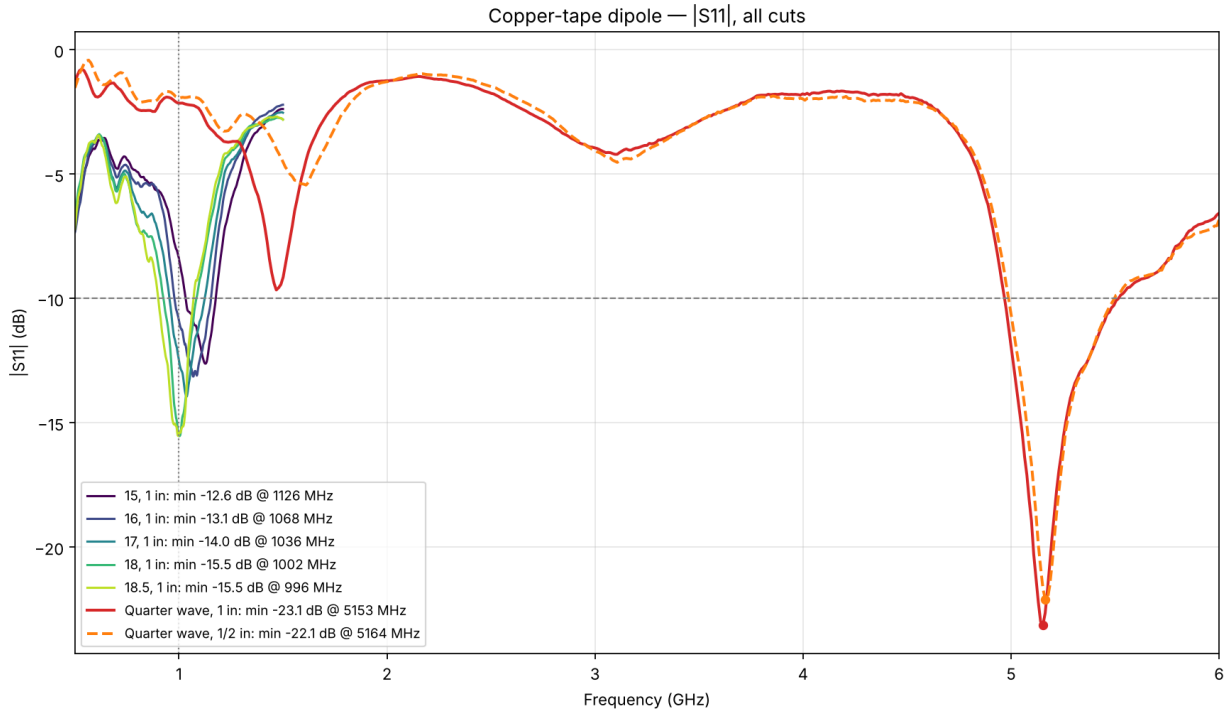


Figure 8: All measured cuts including the quarter-wave specimens (1 in and 0.5 in) to 6 GHz. *[Describe the quarter-wave configuration.]*

6 Comparison with Theory (Exercise 1.6)

[Analytical R_r from the thin-dipole current distribution at each measured length; compare with simulated and measured R_{in}.]

Table 5: Input resistance: theory, simulation, measurement.

L (cm)	R_r analytical (Ω)	R_{in} CST (Ω)	R_{in} measured (Ω)
15	*[...]*	*[...]*	72.1
16	*[...]*	*[...]*	70.2
17	*[...]*	*[...]*	72.0
18	*[...]*	*[...]*	68.4
18.5	*[...]*	*[...]*	65.5

Question 9. How does the analytical radiation resistance compare with the simulated input resistance?

Answer. *[Answer here.]*

Question 10. How closely does the measured resistance agree with simulation?

Answer. *[Answer here.]*

Question 11. What physical effects could account for differences?

Answer. *[Answer here.]*

Question 12. Can the measured input resistance alone establish the antenna's radiation efficiency? Explain.

Answer. *[Answer here.]*

7 Discussion

Resonant length vs. half wavelength

[Answer here.]

What determines impedance bandwidth

[Answer here.]

Copper tape vs. round wire

[Answer here.]

Feed-cable influence on a balanced antenna

[Answer here.]

VNA accuracy and de-embedding

[Answer here.]

Analytical theory vs. numerical simulation

[Answer here.]

8 Conclusion

[Answer here.]